

WAZUH Home Lab - SIEM and File Integrity Monitoring

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In this home lab I set up Wazuh, a free and open-source SIEM and XDR platform, to monitor a Windows machine from a centrally managed Ubuntu server. The goal was to get hands-on experience with how security events and logs are collected, forwarded, and visualized in a real security operations environment.

The lab walks through installing and configuring the Wazuh Manager on an Ubuntu VM, deploying the Wazuh Agent on a Windows host, connecting the two over a local network, and demonstrating File Integrity Monitoring (FIM), where any file created or deleted inside a monitored folder triggers a real-time alert on the dashboard.

Key Skills:

- Log analysis and aggregation
- File Integrity Monitoring (FIM)
- Intrusion detection
- Vulnerability detection
- Real-time alerting and dashboards

Tools Used

- Wazuh - open-source SIEM/XDR platform (manager, agent, dashboard)
- Ubuntu Server 20.04+ - hosts the Wazuh Manager inside VirtualBox
- VirtualBox - hypervisor used to run the Ubuntu VM
- Windows - host machine running the Wazuh Agent
- ossec.conf - Wazuh agent configuration file for enabling FIM

The key points were:

- Installed and configured the Wazuh Manager on an Ubuntu VirtualBox VM
- Accessed and explored the Wazuh web dashboard
- Installed and registered a Wazuh Agent on a Windows host
- Configured File Integrity Monitoring for a specific directory
- Verified real-time alert forwarding from the agent to the manager dashboard

1. Lab Architecture

Component	Host	Role
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Wazuh Manager	Ubuntu (VirtualBox VM)	Collects, analyzes, and stores data from agents. Hosts the web dashboard.
Wazuh Agent	Windows (host machine)	Sends logs, events, and FIM alerts to the Wazuh Manager.

1.1 Same Subnet

WINDOWS

ipconfig

```
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

192.168.56.1

UBUNTU

ifconfig

```
mtran@Ubuntuvm:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.68 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe87:7509 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:87:75:09 txqueuelen 1000 (Ethernet)
    RX packets 375 bytes 27235 (27.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 81 bytes 10259 (10.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Installing the Wazuh Manager (Ubuntu)

I Ran all of the following commands on the Ubuntu VM terminal.

2.1 Add the Wazuh GPG Key

This verifies the authenticity of Wazuh packages before installing them:

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o
/usr/share/keyrings/wazuh-archive-keyring.gpg
```

```
mtran@Ubuntuvm:~$ curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gp
g --dearmor -o /usr/share/keyrings/wazuh-archive-keyring.gpg
```

2.2 Download and Run the Installation Script

This single command downloads and runs the all-in-one installer, setting up the manager, indexer, dashboard, and Filebeat automatically:

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a -i
```

The installation took around 7 minutes. When it finished, the terminal displayed the dashboard URL, username, and auto-generated password. saved all information.

```
mtran@Ubuntuvm:~$ curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a -i
27/03/2026 18:54:51 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: Fm8D.sbQPZXiNtXhEe?aYatKMow+cXD7
```

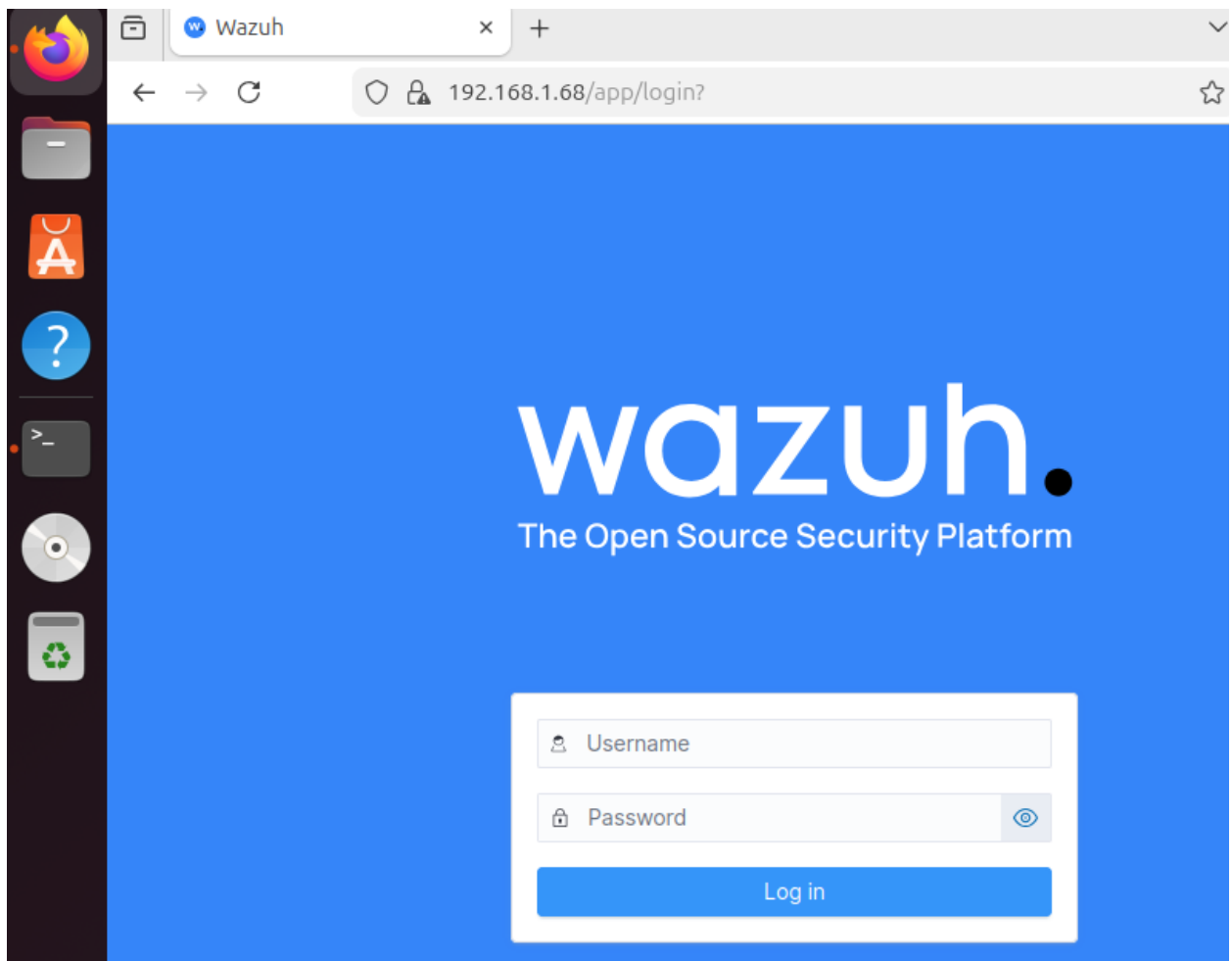
User: admin

Password: Fm8D.sbQPZXiNtXhEe?aYatKMow+cXD7

3. Accessing the Wazuh Dashboard

Once installation was complete, I brought back the Ubuntu VM's IP address:

192.168.1.68 and pasted it in Firefox.



← → ↻ 192.168.1.68/app/server-apis#/settings?tab=api ☆ ⓘ ⚙

≡ W. Server APIs a ⓘ

API Connections

⊕ Add API connection ↻ Refresh ↻ Check updates ⓘ ☐ Disable updates notifications ⓘ

From here you can manage and configure the API entries. You can also check their connection and status.

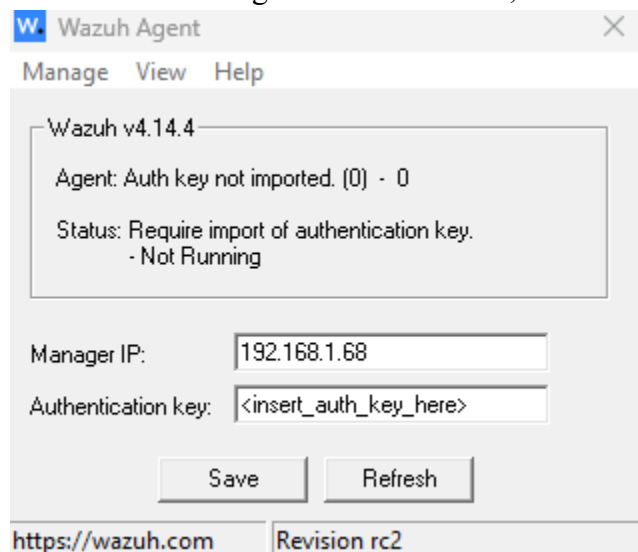
🔍 Search...

ID	Cluster	Manager	Host	Port	Username	Status	Version	Updates status	Run as	Actions
default	Disable	Ubuntuv	https://1	550	wazuh-	● Online	v4.12.0	● Available updates	👁	- ★ ↻

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4. Installing the Wazuh Agent (Windows Host)

1. Downloaded the latest Wazuh Agent MSI installer from the official Wazuh documentation page for Windows.
<https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-windows.html>
2. Ran the MSI installer using default settings.
3. When the installation was completed, I opened the Wazuh Agent Manager from the Start Menu.
4. In the Manager IP Address field, I entered the IP address of the Ubuntu VM.



5. Registering the Agent with the Manager

5.1 Generate the Agent Key on the Ubuntu Manager

On the Ubuntu VM, run the agent management utility

```
sudo /var/ossec/bin/manage_agents
```

Follow the interactive menu:

- Select A to add a new agent
- Assign a name, WAgent
- Enter the Windows machine's IP address - 192.168.56.1
- Confirm to create, it will be assigned an ID
- Select E to extract the key, then enter the agent ID
- Copy the full key string that is displayed

```

* Wazuh v4.12.0 Agent manager.          *
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
Please provide the following:
  * A name for the new agent: WAgent
  * The IP Address of the new agent: 192.168.56.1
Confirm adding it?(y/n): y
Agent added with ID 001.

*****
* Wazuh v4.12.0 Agent manager.          *
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: E

Available agents:
  ID: 001, Name: WAgent, IP: 192.168.56.1
Provide the ID of the agent to extract the key (or '\q' to quit): 001

Agent key information for '001' is:
MDAxIFdBZ2VudCAxOTIuMTY4LjU2LjEgYmU4OWQ2ZWZkMWU0MDAyY2Y5OWViMGM4NDJjNmJjZTIwMjY5ZjYyOTE
4YzhZjliMjdjYTY5YTEwMmRlNWFnNQ==

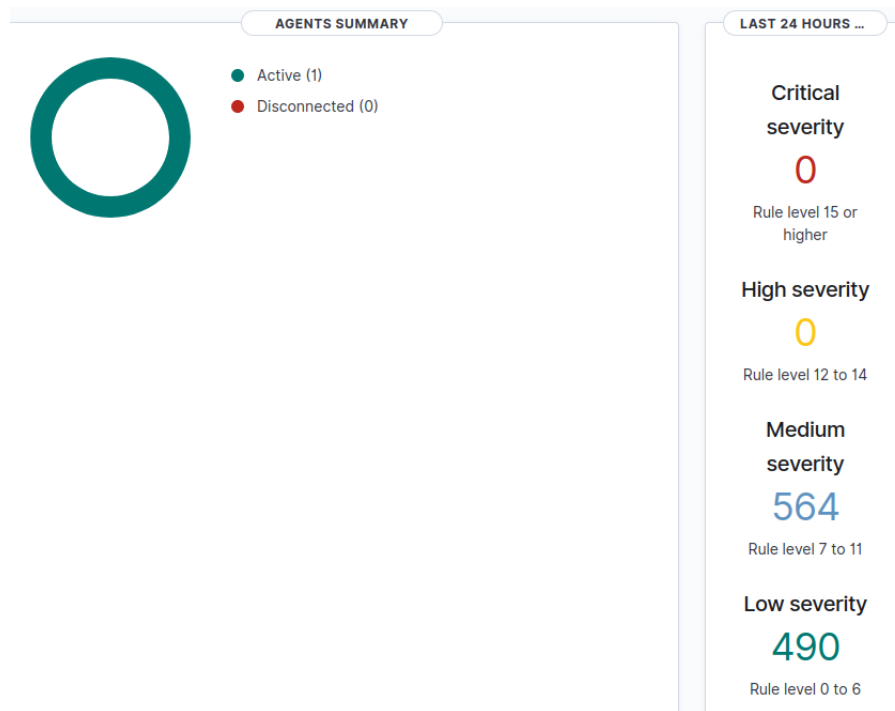
```

MDA1IFdBZ2VudCAxOTIuMTY4LjU2LjEgYmU4OWQ2ZWZkMWU0MDAyY2Y5OWViMGM4NDJjNmJjZTIwMjY5ZjYyOTE4YzhZjliMjdjYTY5YTEwMmRlNWFnNQ==

5.2 Apply the Key on the Windows Agent

5. Open the Wazuh Agent Manager GUI on Windows.
6. Paste the copied key into the Authentication Key field.
7. Save and apply the key.
8. Go to Manage > Restart to restart the agent service.

Go back to the Wazuh dashboard. The Agent Summary should now show one active agent. Click it to confirm your Windows machine is listed with its OS and IP details.



Agents (1)

[+ Deploy new agent](#)

[🔄 Refresh](#)

[📄 Export formatted](#)

[More ▾](#)



☒ Show only outdated

status=active									WQL
☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	005	WAgent	192.168.1.212	default	Microsoft Windows 11 Home 10.0.26200.8039	node01	v4.12.0	● ⓘ	👁️ ⋮

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6. File Integrity Monitoring (FIM)

Wazuh's Syscheck module monitors a directory in real time. Any file created, modified, or deleted inside the folder sends an alert to the Wazuh Manager dashboard.

6.1 Creating a Test Folder on Windows

Create a new folder on the Windows machine to use as the monitored directory:

C:\Users\Matth\Wazuh

6.2 Editing the Agent Configuration File

Open Notepad as Administrator, then open the following file:

C:\Program Files (x86)\ossec-agent\ossec.conf

Find the File Integrity Monitoring (syscheck) section. Inside the directories block, add a new line pointing to your test folder:

```
<directories realtime="yes">C:\Users\YourName\WazuhTest</directories>
```

Save the file with Ctrl+S.

```
<directories realtime="yes">%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup</directories>
<directories realtime="yes">C:\Users\Matth\Wazuh</directories>
```

6.3 Restart the Agent

Restart the Wazuh agent service to apply the configuration change: Manage > Restart in the Wazuh Agent Manager GUI.

7. Verifying the Setup

7.1 Open the Wazuh Dashboard and go to the Windows agent's Integrity Monitoring section.

7.2 On Windows, create a new text file inside the monitored folder

7.3 Refresh the dashboard, a file creation alert appeared with the filename and agent details.

7.4 Delete the file, then refresh again, a file deletion alert appeared.

▼	syscheck.path	▼	syscheck.event	▼	rule.des...	▼	rule.level	▼	rule.id	▼
	c:\users\matth\wazuh\test.txt		deleted		File deleted.		7		553	
	c:\users\matth\wazuh\test.txt		added		File added ...		5		554	
	c:\users\matth\wazuh\new text document.txt		deleted		File deleted.		7		553	
	c:\users\matth\wazuh\new text document.txt		added		File added ...		5		554	

8. Summary.1

This lab demonstrated a complete Wazuh SIEM deployment from scratch, covering potential SOC work. The key points were:

- Installed and configured the Wazuh Manager on an Ubuntu VirtualBox VM
- Accessed and explored the Wazuh web dashboard
- Installed and registered a Wazuh Agent on a Windows host
- Configured File Integrity Monitoring for a specific directory
- Verified real-time alert forwarding from the agent to the manager dashboard

9. Simulated Attack Detection with MITRE ATT&CK Mapping

To extend the lab beyond basic monitoring, I added a Kali Linux attacker machine to my VirtualBox environment, simulated real adversary techniques against the Windows host, and verified that Wazuh detected and categorized them using MITRE ATT&CK framework identifiers. This section documents each step I took, including the exact commands I ran and what I observed in the dashboard.

9.1 Updated Lab Architecture

With the Kali VM added, my lab now consists of three machines on the same Host-Only network:

Component	Host	Role
Wazuh Manager	Ubuntu (VirtualBox VM)	SIEM - collects and analyzes all alerts

Wazuh Agent	Windows (host machine)	Forwards logs and events to the manager
Attacker Machine	Kali Linux (VirtualBox VM)	Used to simulate adversary techniques

9.2 Setting Up the Kali Linux VM

After booting into Kali, I opened a terminal and ran the following command to find the Kali VM's IP address and confirm it was on the same subnet as my other machines:

`ip a`

The output showed a 192.168.x.x address on the Host-Only adapter, confirming Kali was on the same network as the Ubuntu Wazuh Manager and the Windows host.

```

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixro
ute eth0
        valid_lft 547sec preferred_lft 547sec
    inet6 fe80::bab4:55b6:f858:997c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

9.3 Enabling Windows Login Auditing

Before running any attack, I needed to make sure Windows was writing login events to the Security event log. By default, Windows does not always log failed login attempts, so I enabled audit policy manually.

On the Windows machine, I opened the Start Menu and searched for my command prompt and opened it with administrator. Inside the policy editor, I entered the following commands

auditpol /set /subcategory:"Logon" /success:enable /failure:enable

auditpol /set /subcategory:"Account Logon" /success:enable /failure:enable

and to make sure it was what I wanted I used **auditpol /get /category:*** to verify

```
C:\Windows\System32>auditpol /set /category:"Account Logon" /success:enable /failure:enable
The command was successfully executed.

C:\Windows\System32>auditpol /set /category:"Logon/Logoff" /success:enable /failure:enable
The command was successfully executed.

C:\Windows\System32>auditpol /get /category:*
System audit policy
Category/Subcategory      Setting
System
  Security System Extension  No Auditing
  System Integrity           Success and Failure
  IPsec Driver               No Auditing
  Other System Events        Success and Failure
  Security State Change      Success
Logon/Logoff
  Logon                      Success and Failure
  Logoff                     Success and Failure
  Account Lockout            Success and Failure
  IPsec Main Mode            Success and Failure
  IPsec Quick Mode           Success and Failure
  IPsec Extended Mode        Success and Failure
  Special Logon              Success and Failure
  Other Logon/Logoff Events   Success and Failure
  Network Policy Server      Success and Failure
  User / Device Claims        Success and Failure
  Group Membership           Success and Failure
  Access Rights               Success and Failure
Object Access
  File System                No Auditing
  Registry                   No Auditing
  Kernel Object              No Auditing
  SAM                        No Auditing
  Certification Services     No Auditing
  Application Generated       No Auditing
  Handle Manipulation         No Auditing
  File Share                  No Auditing
  Filtering Platform Packet Drop No Auditing
  Filtering Platform Connection No Auditing
  Other Object Access Events   No Auditing
  Detailed File Share          No Auditing
  Removable Storage           No Auditing
  Central Policy Staging       No Auditing
Privilege Use
  Non Sensitive Privilege Use No Auditing
  Other Privilege Use Events   No Auditing
  Sensitive Privilege Use      No Auditing
Detailed Tracking
  Process Creation            No Auditing
  Process Termination          No Auditing
  DPAPI Activity              No Auditing
  RPC Events                  No Auditing
  Plug and Play Events         No Auditing
  Token Right Adjusted Events   No Auditing
Policy Change
  Audit Policy Change          Success
  Authentication Policy Change Success
  Authorization Policy Change No Auditing
  MPSSVC Rule-Level Policy Change No Auditing
  Filtering Platform Policy Change No Auditing
  Other Policy Change Events    No Auditing
Account Management
  Computer Account Management No Auditing
  Security Group Management    Success
  Distribution Group Management No Auditing
  Application Group Management No Auditing
  Other Account Management Events No Auditing
  User Account Management      Success
DS Access
  Directory Service Access     No Auditing
  Directory Service Changes     No Auditing
  Directory Service Replication No Auditing
  Detailed Directory Service Replication No Auditing
Account Logon
  Kerberos Service Ticket Operations Success and Failure
  Other Account Logon Events    Success and Failure
  Kerberos Authentication Service Success and Failure
  Credential Validation          Success and Failure
```

9.4 Running the Brute Force Attack from Kali

With auditing enabled on Windows, I switched to the Kali VM and opened a terminal. I used Hydra, a credential brute force tool that is pre-installed on Kali, to simulate a password attack against the Windows machine over SMB (Server Message Block), which is the protocol Windows uses for file and authentication sharing on a network.

I first make sure the rockyou wordlist is unzipped. On Kali it sometimes comes compressed.

```
gunzip /usr/share/wordlists/rockyou.txt.gz
```

I ran the following command, replacing the IP address at the end with the actual IP of my Windows host:

```
hydra -l administrator -P /usr/share/wordlists/rockyou.txt 192.168.56.1 smb
```

This command told Hydra to target the administrator account and cycle through the rockyou.txt wordlist, which is a large list of commonly used passwords included with Kali. Hydra began attempting each password one by one against the Windows SMB login.

I let the attack run for about 45 seconds to generate enough failed login events in the Windows Security log, then stopped it with Ctrl+C. The attack did not need to succeed. The goal was purely to generate the failed authentication events that Wazuh would detect.

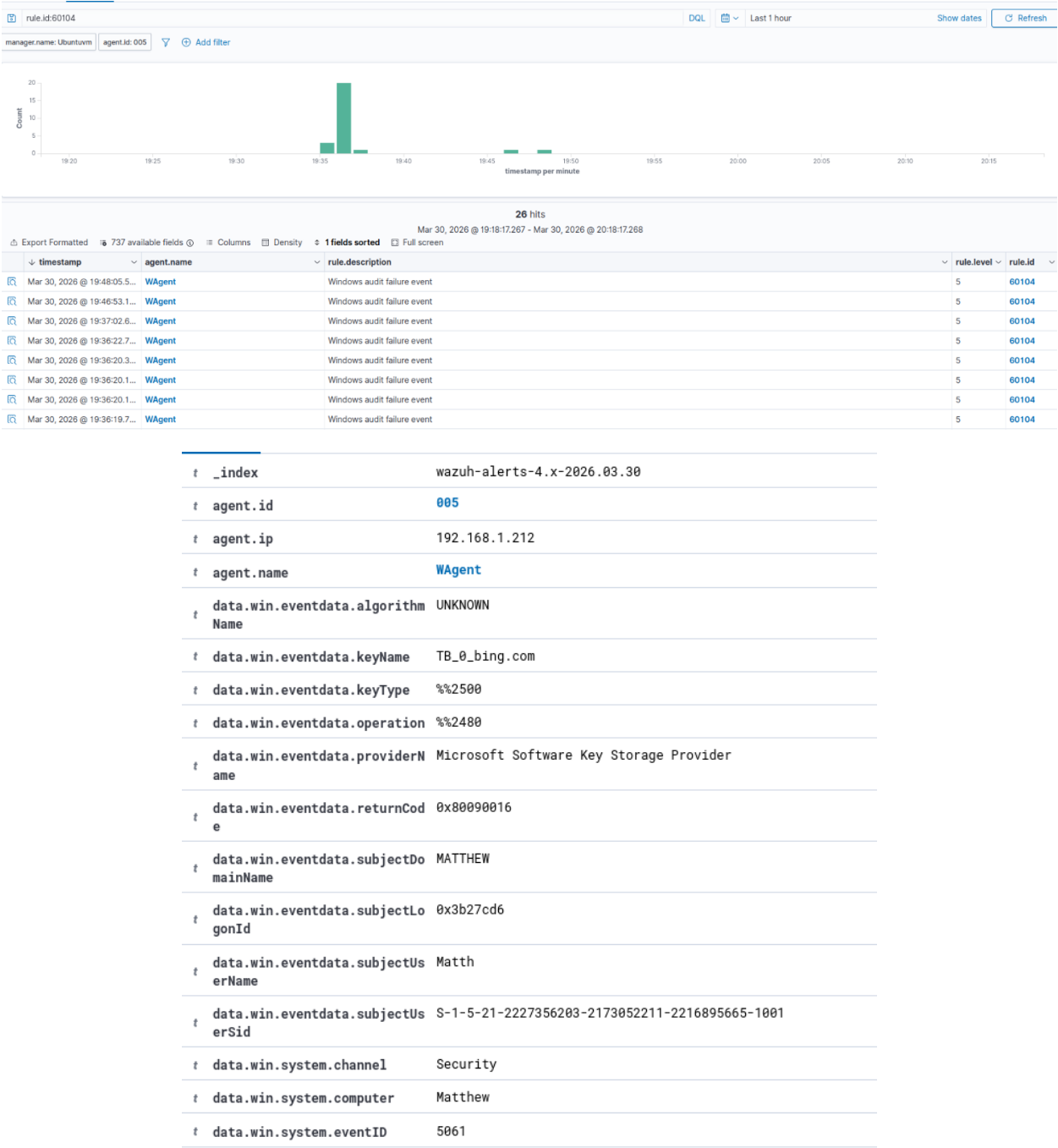
```
(kali㉿kali)-[~]
$ hydra -l administrator -P /usr/share/wordlists/rockyou.txt 192.168.56.1 s
mb
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-30 19:
49:13
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[DATA] max 1 task per 1 server, overall 1 task, 14344399 login tries (l:1/p:1
4344399), ~14344399 tries per task
[DATA] attacking smb://192.168.56.1:445/
C^[[B^C
```

9.5 Verifying the Alerts in Wazuh

After stopping Hydra, I opened the Wazuh dashboard in Firefox on the Ubuntu VM. I navigated to the threat hunting section. I could immediately see a large spike of alerts in the timeline

corresponding to the time I ran Hydra. I also searched for rule.id:60104, which is Wazuh's built-in rule for Windows audit failure.



9.6 Simulating Reconnaissance with Nmap

While still on the Kali VM, I also ran a basic port scan against the Windows machine to simulate a reconnaissance technique. I opened a terminal and ran:

`nmap -sS 192.168.56.1`

```
(kali@kali)-[~]  
$ nmap -sS 192.168.56.1  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-30 20:34 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00036s latency).  
All 1000 scanned ports on 192.168.56.1 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
MAC Address: 0A:00:27:00:00:0B (Unknown)  
  
Nmap done: 1 IP address (1 host up) scanned in 23.97 seconds
```

The `-sS` flag performs a SYN scan, which is a common stealth scanning technique used by attackers to map open ports on a target without completing a full TCP connection. This technique maps to MITRE ATT&CK T1046 - Network Service Discovery.

After the scan was completed, I checked the Wazuh dashboard for corresponding alerts. No alert fired for the port scan. This is a detection gap. Wazuh's default ruleset detected the brute force authentication failures but did not generate an alert for the network reconnaissance activity.